



**SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)**

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)  
Re-Accredited by NAAC with 'A++' Grade

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Batch : 2025-29

**Assignment : 4**

**Subject: Programming with Java**

**ASSIGNMENT No : 4**

**TITLE :**Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

**Problem Statement :** Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

**Learning Objectives :** To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

**Theory :** Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

**Conclusion :** The experiment demonstrated the use of Java wrapper classes for converting between String, primitive data types, and wrapper objects. It also showed how converted values can be validated and used to perform basic arithmetic operations, improving data handling and program reliability.



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## Output Screenshots

```
J WrapperExample.java > Java > WrapperExample
1 class WrapperExample {
    Run main | Debug main | Run | Debug
2     public static void main(String[] args) {
3         int x = 512;
4         Integer wrappedX = Integer.valueOf(x);
5         int unwrappedX = wrappedX.intValue();
6         System.out.println("Wrapped : " + wrappedX + ", Unwrapped: " + unwrappedX);
7     }
8 }
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\LENOVO\Documents\JAVA Assignments> javac WrapperExample.java
- PS C:\Users\LENOVO\Documents\JAVA Assignments> java WrapperExample  
Wrapped : 512, Unwrapped: 512
- PS C:\Users\LENOVO\Documents\JAVA Assignments>



### Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
J Swrap.java > Language Support for Java(TM) by Red Hat > Swrap > main(String[])
1  public class Swrap {
2      String marks;
3
4      Swrap(String m) {
5          marks = m;
6      }
7      void displaystr() {
8          System.out.println("Marks (string): " + marks);
9      }
10     void displayint() {
11         int marksInt = Integer.parseInt(marks);
12         System.out.println("Marks (int): " + marksInt);
13     }
14     Run main | Debug main | Run | Debug
15     public static void main(String[] args) {
16         Swrap s1 = new Swrap("67");
17         s1.displaystr();
18         s1.displayint();
19     }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Swrap.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java Swrap
Marks (string): 67
Marks (int): 67
❖ PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Ans) a) Case 1 :

```
J employee.java > Language Support for Java(TM) by Red Hat > Employee
1  public class Employee {
2      String ID;
3      String bsal;
4      String bonus;
5
6      int intID;
7      int intbsal;
8      int intbonus;
9
10     Employee(String ID, String bsal, String bonus) {
11         this.ID = ID;
12         this.bsal = bsal;
13         this.bonus = bonus;
14     }
15     void stoint() {
16         intID = Integer.parseInt(ID);
17         intbsal = Integer.parseInt(bsal);
18         intbonus = Integer.parseInt(bonus);
19     }
20     int validsal() {
21         if (intbsal < 0 || intbonus < 0) {
22             System.out.println("Invalid salary or bonus.");
23             return 0;
24         } else {
25             System.out.println("Valid salary and bonus.");
26             return 1;
27         }
28     }
29     void totalsal() {
30         int totalSalary = intbsal + intbonus;
31         if(validsal() == 0) {
32             System.out.println("Cannot calculate total salary due to invalid input.");
33             return;
34         }
35         System.out.println("Total Salary calculated from basic and bonus " + totalSalary);
36     }
37     Run main | Debug main | Run | Debug
38     public static void main(String[] args) {
39         Employee e1 = new Employee("055", "75000", "5000");
40         e1.stoint();
41         e1.totalsal();
42     }
43 }
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Employee.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java Employee
Valid salary and bonus.
Total Salary calculated from basic and bonus 80000
❖ PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



### b) Case 2 :

```
J employee.java > Java > Employee > main(String[] args)
1  public class Employee {
2      String ID;
3      String bsal;
4      String bonus;
5
6      int intID;
7      int intbsal;
8      int intbonus;
9
10     Employee(String ID, String bsal, String bonus) {
11         this.ID = ID;
12         this.bsal = bsal;
13         this.bonus = bonus;
14     }
15     void stoint() {
16         intID = Integer.parseInt(ID);
17         intbsal = Integer.parseInt(bsal);
18         intbonus = Integer.parseInt(bonus);
19     }
20     int validsal() {
21         if (intbsal < 0 || intbonus < 0) {
22             System.out.println("Invalid salary or bonus.");
23             return 0;
24         } else {
25             System.out.println("Valid salary and bonus.");
26             return 1;
27         }
28     }
29     void totalsal() {
30         int totalSalary = intbsal + intbonus;
31         if(validsal() == 0) {
32             System.out.println("Cannot calculate total salary due to invalid input.");
33             return;
34         }
35         System.out.println("Total Salary calculated from basic and bonus " + totalSalary);
36     }
37     Run main | Debug main | Run | Debug
38     public static void main(String[] args) {
39         Employee e1 = new Employee("055", "-23300", "3000");
40         e1.stoint();
41         e1.totalsal();
42     }
43 }
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Employee.java
- PS C:\Users\LENOVO\Documents\JAVA Assignments> java Employee  
Invalid salary or bonus.  
Cannot calculate total salary due to invalid input.
- ❖ PS C:\Users\LENOVO\Documents\JAVA Assignments>